

ValuJet Article Summary

The article starts off with an eyewitness account of an airplane crash in 1996 by Walton Little, setting up the subsequent context and background for the rest of the article. William Langewiesche goes into further detail about the different types of aircraft accidents: procedural, engineered, and system accidents, with the ValuJet crash being the latter. He then provides a play-by-play of what happened during the accident, including there was mass passenger panic as smoke and fire spread through the main cabin, the subsequent miscommunication between the pilots and air traffic controllers, and the failure to recognize the immediate danger due to the rapidly developing fire, toxic smoke, and lack of time to respond. Langewiesche also digs into how ValuJet, the operating airline, worked to keep costs as low as possible, including through outsourcing and a low-cost operating model. As subsequent investigations discovered through wreckage recovery, the failure was not simply a one-off aircraft problem but an issue originating in the system as a whole. ValuJet opted to outsource work whenever possible to cut costs, leading to a series of safety-check failures that contributed to the deaths of 110 passengers and crew members. Oxygen generators that could produce intense heat when activated were removed from aircraft but were not properly capped or stored. The mechanics failed to follow the written instructions to install safety caps, and some work orders were falsely signed off. The generators were then moved into the shipping area, where they were mistakenly treated as ordinary cargo and eventually returned to ValuJet. The

problem was further complicated by confusing terminology surrounding whether the generators were "expired" or "expended," making it difficult for workers to understand which generators were still capable of activation. Each individual failure appeared relatively minor, but as they accumulated and interacted with one another, they ultimately contributed to the crash. There was much public outcry due to the devastating nature of the accident, combined with the lack of a tactful response from ValuJet and its leadership, and blame and responsibility were demanded. However, with such a large systemic problem, it became difficult to pinpoint where the root of the error lay. It could have been the lack of clear instructions, confusing and tedious safety procedures, paperwork that did not accurately reflect what had actually happened, or individual employees becoming accustomed to overlooking rules. David Hinson, the head of the FAA, did not accept responsibility despite evidence pointing toward multiple FAA concerns over ValuJet's alarming growth in conjunction with its lack of quality and safety oversight. In the end, ValuJet was grounded, while SabreTech, the outsourced maintenance company, lost customers and eventually closed its Miami operation.

Langewiesche then discusses the broader implications of such an accident through Charles Perrow's theory of the "normal accident." Perrow argues that increasingly complex systems can produce accidents through unexpected combinations of otherwise ordinary failures. In tightly coupled systems, these failures can interact and escalate faster than people are able to recognize or correct them.

Langewiesche argues that the ValuJet crash represents a nearly

perfect example of a system accident because the fire was not caused by one isolated mistake, but by a chain of interconnected failures across the airline, maintenance company, shipping process, and regulatory system.

From the story of the ValuJet accident, system-accident thinking can be applied to the software engineering process to create software that is fault-tolerant. This is important because as a software scales in components and users it should respond to the possible fail cases. However, one of the main takeaways of the article is that every possible fail case cannot be possibly thought of so proper systems must be put in place to verify and test for failures. The article described the three types of accidents as procedural, engineered and systemic. From this article, an application to software engineering can be to use procedural and engineered accidents to analyze and change the structure in which the software was produced. For example if a pilot is sleep-deprived, then it should be questioned about the scheduling system that has overworked the pilot. In addition to this problem, an engineering solution could be to create a scheduling system that allows a pilot to get proper sleep or detect when a pilot is showing sleep-deprived behaviors. In the article it also mentions that the ValuJet company had a business model to cut costs as much as possible by providing lower salaries to their employees. Given this information, an application to software engineering would be to change the goal from making low-cost decisions to making customer-focused decisions.

However there are always the pressures of the business model which must be balanced with the goal of keeping customers safe.